

Probabilistic Reasoning Over Time

1 Markov Chains

Stochastic Process A stochastic process is a family of random variables $\{X(t) : t \in T\}$ defined on a *probability space*, where T is the *parameter set*. When $T = \{0, 1, 2, 3, \dots\}$, it is usually referred to as a time series. When T is a vector space, the stochastic process is called a random field. The collection of all possible states (values) of $X(t)$ forms the *state space*, which can be either discrete or continuous.

Markov Assumption / Markov Property; Markov Process / Markov Chain A Markov process (or Markov chain) is a stochastic process indexed by time, which satisfies the Markov assumption (or Markov property) — namely, the *conditional probability distribution* of the current state depends on only a *finite fixed number* of previous states, i.e.,

$$P(X_t | \{X_{t-1}, X_{t-2}, \dots\}) = P(X_t | \{X_{t-1}, X_{t-2}, \dots, X_{t-k}\}) \quad \text{for any } t$$

First-Order Markov Process A first-order Markov process is a Markov process where the current state depends only on the immediately preceding state, i.e.,

$$P(X_t | \{X_{t-1}, X_{t-2}, \dots\}) = P(X_t | X_{t-1}) \quad \text{for any } t$$

In the following sections, we will focus on first-order Markov processes.